

Wilson loop expectations for finite gauge groups

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Introduction

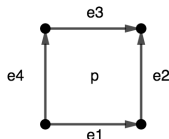
- ▶ Lattice gauge theories are models from physics obtained by discretizing continuous spacetime $\mathbb{R}^4$ by a lattice $\varepsilon\mathbb{Z}^4$.
- ▶ They are rigorously defined, so we can actually prove things.
- ▶ The continuum counterparts to lattice gauge theories are Euclidean Yang-Mills theories. Rigorous construction of such theories is of physical importance.
- ▶ One approach to rigorously defining a Euclidean Yang-Mills theory is to take a scaling limit of lattice gauge theories.
- ▶ In order to do so, various properties of lattice gauge theories must be very well understood.
- ▶ This talk is about understanding one particular property.

Wilson loops

- ▶ The key objects of interest in a lattice gauge theory are the Wilson loops, which are certain random variables.
- ▶ Introduced by Wilson (1974) to give a theoretical explanation of an observed phenomenon.
- ▶ The explanation was in terms of Wilson loop expectations.
- ▶ Since then, there have been a number of results analyzing Wilson loop expectations; see Chatterjee's survey "Yang-Mills for probabilists" for a detailed overview.
- ▶ One approach to taking a scaling limit involves being able to calculate Wilson loop expectations. More on this later.

Notation

- ▶ Let G be a group, whose elements are $d \times d$ unitary matrices. The identity is denoted I_d . We will often refer to G as the gauge group.
- ▶ Let $\Lambda := [-N, N]^4 \cap \mathbb{Z}^4$ be a large box.
- ▶ Let Λ_1 be the set of positively oriented edges in Λ . An edge (x, y) is positively oriented if $y = x + e_i$, for some standard basis vector e_i .
- ▶ Let Λ_2 denote the set of plaquettes in Λ . A plaquette is a unit square whose four boundary edges are in Λ . Pictorially:



- ▶ Given an edge configuration $\sigma : \Lambda_1 \rightarrow G$, and a plaquette p as above, define $\sigma_p := \sigma_{e_1} \sigma_{e_2} \sigma_{e_3}^{-1} \sigma_{e_4}^{-1}$.

Definition of lattice gauge theories

- ▶ Define

$$S_{\Lambda}(\sigma) := \sum_{p \in \Lambda_2} \operatorname{Re}(\operatorname{Tr}(I_d) - \operatorname{Tr}(\sigma_p)).$$

- ▶ Let $G^{\Lambda_1} := \{\sigma : \Lambda_1 \rightarrow G\}$. Let μ_{Λ} be the product uniform measure on G^{Λ_1} .
- ▶ For $\beta \geq 0$, define the probability measure $\mu_{\Lambda, \beta}$ on G^{Λ_1} by

$$d\mu_{\Lambda, \beta}(\sigma) := Z_{\Lambda, \beta}^{-1} \exp(-\beta S_{\Lambda}(\sigma)) d\mu_{\Lambda}(\sigma).$$

- ▶ We say that $\mu_{\Lambda, \beta}$ is the lattice gauge theory with gauge group G , on Λ , with inverse coupling constant β .
- ▶ Examples: $G = U(1), SU(2), SU(3)$.
- ▶ For $U(1)$, the continuum theory may be constructed directly - Gross (1986). Convergence of the lattice $U(1)$ theory was established by Driver (1987).

Definition of Wilson loops

- ▶ Let γ be a closed loop in Λ , with directed edges $e_1, \dots, e_n$.
- ▶ The Wilson loop variable W_γ is defined as

$$W_\gamma(\sigma) := \text{Tr}(\sigma_{e_1} \cdots \sigma_{e_n}).$$

[If e is negatively oriented, then $\sigma_e := \sigma_{-e}^{-1}$.]

- ▶ Let $\langle W_\gamma \rangle_{\Lambda, \beta}$ be the expectation of W_γ under $\mu_{\Lambda, \beta}$. Define

$$\langle W_\gamma \rangle_\beta := \lim_{\Lambda \uparrow \mathbb{Z}^4} \langle W_\gamma \rangle_{\Lambda, \beta}.$$

[This limit may only exist after taking a subsequence, but I will pretend that this technical point is not present.]

A leading order computation

- ▶ Recently, Chatterjee (2018) computed Wilson loop expectations to leading order at large β , when $G = \mathbb{Z}_2 = \{\pm 1\}$.
- ▶ For a loop of length ℓ , we have

$$\langle W_\gamma \rangle_\beta \approx e^{-2\ell e^{-12\beta}}.$$

- ▶ Suppose we have a loop γ of length ℓ in $\mathbb{R}^4$. For $\varepsilon > 0$, we can obtain a discretization γ_ε in $\varepsilon\mathbb{Z}^4$ of length $\varepsilon^{-1}\ell$.
- ▶ If we set $\beta_\varepsilon := -\frac{1}{12} \log \varepsilon$, then as $\varepsilon \downarrow 0$, we have

$$\langle W_{\gamma_\varepsilon} \rangle_{\beta_\varepsilon} \longrightarrow e^{-2\ell}.$$

- ▶ This is the first step in one approach to taking a scaling limit.
- ▶ We thus want to understand the leading order of $\langle W_\gamma \rangle_\beta$ at large β , for general gauge groups.

Main result

- ▶ In recent work, I've computed the leading order for finite gauge groups. First, some notation for the formula.
- ▶ Define

$$\Delta_G := \min_{g \neq I_d} \operatorname{Re}(\operatorname{Tr}(I_d) - \operatorname{Tr}(g)).$$



$$G_0 := \{g \in G : \operatorname{Re}(\operatorname{Tr}(I_d) - \operatorname{Tr}(g)) = \Delta_G\}.$$



$$A := \frac{1}{|G_0|} \sum_{g \in G_0} g.$$

Theorem (C. 2020)

Let $\beta \geq \Delta_G^{-1}(1000 + 14 \log|G|)$. Let γ be a loop of length ℓ . Let $X \sim \operatorname{Poisson}(\ell|G_0|e^{-6\beta\Delta_G})$. Then

$$|\langle W_\gamma \rangle_\beta - \operatorname{Tr}(\mathbb{E} A^X)| \leq 10de^{-c(G)\beta}.$$

Main result (cont.)

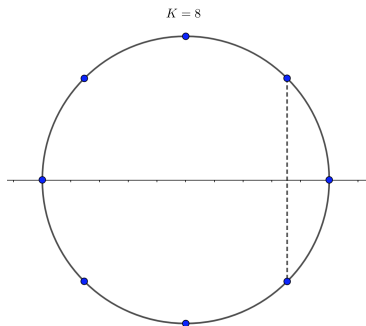
- ▶ Let $-1 \leq \lambda_1, \dots, \lambda_d \leq 1$ be the eigenvalues of A . Then

$$\mathrm{Tr}(\mathbb{E}A^X) = \sum_{i=1}^d e^{-(1-\lambda_i)\ell|G_0|} e^{-6\beta\Delta_G}.$$

- ▶ There is a recent article by Forsström, Lenells, and Viklund (2020), which handles finite Abelian gauge groups. They are able to obtain a much better β threshold in this case.

Main result (cont.)

- ▶ Example: take $K \geq 2$. Let $G = \{e^{i2\pi k/K}, 0 \leq k \leq K-1\}$.



- ▶ Then $\Delta_G = 1 - \cos(2\pi/K)$, $G_0 = \{e^{i2\pi/K}, e^{-i2\pi/K}\}$, and $A = \cos(2\pi/K)$. Letting $\lambda := \ell|G_0|e^{-6\beta(1-\cos(2\pi/K))}$, then

$$\mathbb{E} A^{\text{Poisson}(\lambda)} = e^{-\lambda(1-A)}.$$

- ▶ If $K = 2$, then $\Delta_G = 2$, $G_0 = \{-1\}$, $A = -1$, $\lambda = \ell e^{-12\beta}$.

Rest of the talk

- ▶ I will first outline the proof of the theorem in the Abelian case. The main probabilistic insights are already all present.
- ▶ In essence, the proof has two main steps.
 - Use a Peierls-type argument to show $\langle W_\gamma \rangle_\beta \approx \text{Tr}(\mathbb{E} A^{N_\gamma})$, where N_γ is a count of weakly dependent rare events.
 - Show $N_\gamma \approx \text{Poisson}$.
- ▶ This two step outline was already present in Chatterjee (2018).
- ▶ When the gauge group is non-Abelian, this still works. I will describe the main ideas behind showing this.

Preliminaries

- ▶ Suppose $d = 1$. Take some large box Λ .
- ▶ Recall $\mu_{\Lambda,\beta}$ is a probability measure on G^{Λ_1} with the form

$$\mu_{\Lambda,\beta}(\sigma) = Z_{\Lambda,\beta}^{-1} \exp\left(-\beta \sum_{p \in \Lambda_2} (1 - \operatorname{Re}(\sigma_p))\right).$$

- ▶ Define

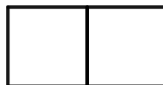
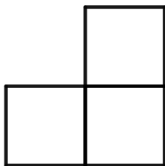
$$\operatorname{supp}(\sigma) := \{p \in \Lambda_2 : \sigma_p \neq 1\}.$$

Let $\Sigma \sim \mu_{\Lambda,\beta}$. Let $\mathbf{S} := \operatorname{supp}(\Sigma)$.

- ▶ When β is large, $\Sigma_p = 1$ for most p , and so $\mathbf{S}$ is typically composed of sparsely distributed clumps.
- ▶ We will see that $\mathbf{S}$ is easier to work with than Σ .

Picture to keep in mind

- ▶ Here's a 2D cartoon of **S**:



Decomposing $\mathbf{S}$

- ▶ Since $\mathbf{S}$ is typically made of sparsely distributed clumps, let us try to decompose $\mathbf{S}$ into more elementary components.
- ▶ In general, any $P \subseteq \Lambda_2$ has a unique decomposition $P = V_1 \cup \dots \cup V_n$ into “connected components”.

Definition

A set $V \subseteq \Lambda_2$ is called a **vortex** if it cannot be decomposed further.

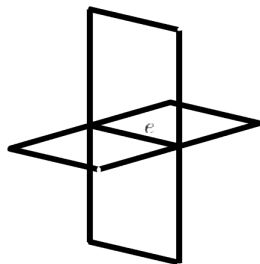
- ▶ It turns out that if $P = V_1 \cup \dots \cup V_n$, then

$$\mathbb{E}[W_\gamma(\Sigma) \mid \mathbf{S} = P] = \prod_{i=1}^n \mathbb{E}[W_\gamma(\Sigma) \mid \mathbf{S} = V_i].$$

- ▶ So we want to understand $\mathbb{E}[W_\gamma(\Sigma) \mid \mathbf{S} = V]$, for vortices V .

Understanding vortex contributions

- ▶ For a vortex V , we say that V appears in $\mathbf{S}$ if V is in the vortex decomposition of $\mathbf{S}$.
- ▶ For an edge e , define $P(e)$ to be the set of plaquettes which contain e . Note $|P(e)|= 6$.



- ▶ In 3D, $P(e)$ looks like this.
- ▶ The smallest vortex which can appear in $\mathbf{S}$ must be $P(e)$, for some edge e . All other vortices must have size ≥ 7 .

Understanding vortex contributions (cont.)

- ▶ For a vortex $P(e)$, we have

$$\mathbb{E}[W_\gamma(\Sigma) \mid \mathbf{S} = P(e)] = \begin{cases} 1 & e \notin \gamma \\ A & e \in \gamma \end{cases}.$$

- ▶ Let N_γ be the number of edges e in γ such that $P(e)$ appears in $\mathbf{S}$. We then have

$$\mathbb{E}[W_\gamma(\Sigma) \mid \mathbf{S}] = A^{N_\gamma} Y,$$

where Y is the contribution from vortices of size ≥ 7 .

- ▶ Next, we show that with high probability, we can ignore vortices of size ≥ 7 .

Understanding vortex contributions (cont.)

- ▶ In order for a vortex V to be such that $\mathbb{E}[W_\gamma(\Sigma) \mid \mathbf{S} = V] \neq 1$, it must be close to the loop γ .
- ▶ Larger vortices are much less likely to appear in $\mathbf{S}$.
- ▶ So if we look in a neighborhood of γ , only $P(e)$ vortices are likely to appear.
- ▶ We thus have

$$\mathbb{P}(Y \neq 1) = \text{lower order.}$$

- ▶ Thus on an event of high probability,

$$\mathbb{E}[W_\gamma(\Sigma) \mid \mathbf{S}] = A^{N_\gamma}.$$

Poisson approximation

- ▶ It remains to show $N_\gamma \approx \text{Poisson}$.
- ▶ We apply the dependency graph approach to Stein's method.
- ▶ Given vortices $V_1 = P(e_1), \dots, V_n = P(e_n)$ which are not too close to each other, we need to have

$$\mathbb{P}(V_1, \dots, V_n \text{ appear in } \mathbf{S}) \approx \prod_{i=1}^n \mathbb{P}(V_i \text{ appears in } \mathbf{S}).$$

- ▶ This is done by cluster expansion. Cluster expansion is a fairly well known tool; for example it appears in Seiler's 1982 monograph on lattice gauge theories.
- ▶ Thus we see why $\mathbf{S}$ is nice to work with: it has “a lot of independence”.

The general case

- ▶ There are some technical difficulties that appear in the non-Abelian case.
- ▶ In the remaining time, I will present the key idea needed to handle these difficulties.
- ▶ I will then give a toy example showing why the key idea is useful.

The key idea

- ▶ Let us now think of Λ as a graph.
- ▶ The fundamental group $\pi_1(\Lambda)$ is made of (equivalence classes of) closed loops in Λ which begin and end at some fixed basepoint. Every closed loop is given by some sequence of edges $e_1 \cdots e_n$.

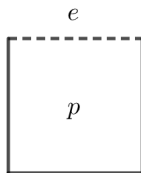
Observation (Szlachànyi and Vecsernyès (1989))

Any $\sigma \in G^{\Lambda_1}$ induces a homomorphism $\psi_\sigma : \pi_1(\Lambda) \rightarrow G$, defined by

$$\psi_\sigma(e_1 \cdots e_n) := \sigma_{e_1} \cdots \sigma_{e_n}.$$

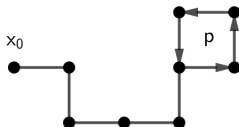
Toy example

- ▶ Let T be a spanning tree of Λ . Suppose $\sigma \in G^{\Lambda_1}$ is such that $\sigma_e = I_d$ for all $e \in T$.
- ▶ Suppose additionally that $\sigma_p = I_d$ for all $p \in \Lambda_2$.
- ▶ I then claim that in fact, $\sigma_e = I_d$ for all $e \in \Lambda_1$.
- ▶ Initial attempt:



Toy example (cont.)

- ▶ The crucial topological fact: any loop in $\pi_1(\Lambda)$ is a product of “Lasso type” loops:



- ▶ For any Lasso type loop $L \in \pi_1(\Lambda)$, we have $\psi_\sigma(L) = I_d$.
- ▶ Thus ψ_σ is trivial.
- ▶ Given $e = (x, y) \in \Lambda_1 \setminus T$, take a loop $a_e \in \pi_1(\Lambda)$ which uses e , and such that every other edge of a_e is in T .
- ▶ Since $\sigma = I_d$ on T , we have $\psi_\sigma(a_e) = \sigma_e$.
- ▶ Because ψ_σ is trivial, $\psi_\sigma(a_e) = I_d$.

In summary

- ▶ Computing the leading order of Wilson loop expectations is a first step in taking a scaling limit.
- ▶ The key probabilistic insight:

$$\mathbb{E}[W_\gamma(\Sigma) \mid \mathbf{S}] = \text{Tr}(A^{N_\gamma}) \quad \text{w.h.p.}$$

- ▶ The key technical tool: the components of $\mathbf{S}$ appear essentially independently.
- ▶ In the non-Abelian case, algebraic topology is a natural language to use.
- ▶ Special thanks to Sourav Chatterjee, whose conversations shaped this work, as well as Persi Diaconis, Hongbin Sun, and Ciprian Manolescu. I am particularly indebted to Ciprian Manolescu for the proof of a key technical lemma.